Lead 4: small-root / algebraic decomposition oracles for S4 and S5 over prime fields

2026-10-28

Lead 4: small-root and algebraic decomposition oracles for S_4/S_5 over F_p

Status. Ideation and paper analysis (§0–§7), followed by measured probes with native code (§7a, §7b). No breakthrough is claimed. No idea below plausibly reaches the S_4 milestone ($\delta \geq 1/4$) or the S_5 goal ($\delta \approx 1/5$). The four ranked records are ordered by expected information per unit cost, not by any chance of success.

Labels used throughout. `[measured]`: a run in this repository, cited. `[derived]`: computed in this note and checkable from the stated formula. `[literature]`: reported in a cited source, with how much of it was read. `[proposed]`: an idea or prediction, not a result. `[recalled]`: from memory, not re-read this session.

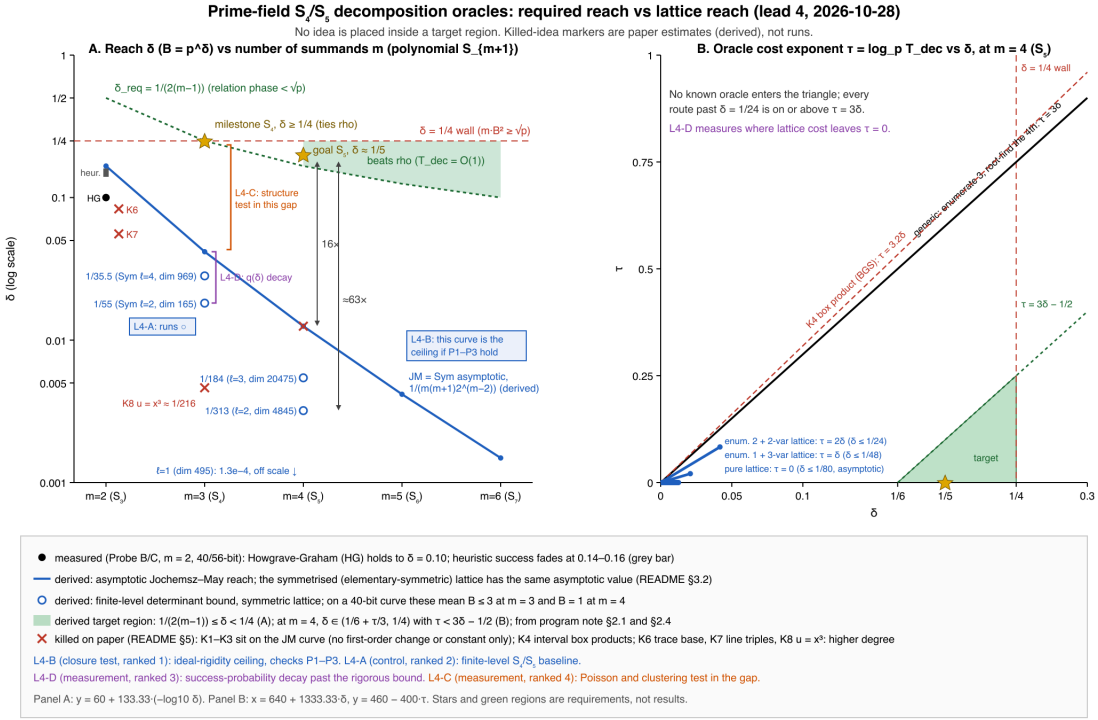


Figure 1: δ-vs-arity map (A) and the m = 4 oracle-cost plane (B). Source: delta_arity_map.svg in this directory.

0. Bottom line

1. [derived] **The "1/80" for S_5 is an asymptotic formula and was never measured.** Probes B and C measured m = 2 (S_3) only. At lattice dimensions that can actually be reduced (about $5 \cdot 10^3$ or less), the determinant bound at m = 4 is $\delta \approx 1/313$ (symmetric lattice, level 2, dimension 4845). On a 40-bit curve that means B = 1. The shortfall to $\delta \approx 1/5$ at feasible dimension is about **63×**, not 16×. At m = 3 the best feasible bound is $\delta \approx 1/35.5$ (dimension 969), which is $B \approx 2$ at 40 bits, against the milestone's $B = 2^{10}$ (§3.2).
2. [derived] **Six structural facts (F1–F6, §3.3) bound every lattice oracle on a coordinate factor base on a prime-order curve.** They are a principal ideal, Newton-polytope rigidity, the degree-optimality of x, box-volume invariance, the cost of overdetermination, and the global-lifting wall. Together they explain why every lattice variant measured so far lands on the same threshold, and they kill the 16 mechanisms in §5.
3. [proposed] **Rank 1 is a closure test (L4-B).** It turns F1–F4 into a checkable statement with three premises, P1–P3. Its first step is symbolic and takes minutes. If a premise fails, that failure is the only lattice lead this note leaves open.
4. [proposed] **The narrow remaining space** (§7) is: a non-lattice small-root algorithm for one structured congruence at box volume p^{0.7–0.8}; factor bases that are not coordinate boxes (implicit, high-degree, or target-dependent); and transfer (§4.3 of the program note). This note has no concrete candidate in any of them.
5. [derived] **A planned experiment looks inapplicable.** The "D_4-symmetrised S_5 lattice probe" in `RESEARCH_IC_LEADS_AND_CONNECTIONS_20261010.md` §8.2 and §9.5 needs a rational 2- or 4-torsion point (as in FHJRV). That point does not exist on a prime-order curve. On even-order test curves the gain is a constant (K3, §5).
6. [measured] **The follow-up probes, run with native code (§7b), find nothing past the bounds.**
   - L4-B step 1 (native port) agrees with GP: S_4 and S_5 are dense and irreducible over F_p.
   - L4-C: an exact census over all targets on six toy curves finds no extra additive structure in small-x factor bases (41/41 cells inside the null band).
   - L4-A: symmetric S_4 lattices at dims 35 and 165 recover no planted non-global root at $B \geq 2$, matching §3.2. A v1 false positive came from global relations and was amended.
   - P3: the ePrint LP could not be retrieved. Our search over triangular shift lattices exceeds no JM asymptote.
   - P3, addendum (2026-10-29, §7c): the strategy-independent bound of `research/smallroot_optimal_reach_20261026` applies here. Any x-variable lattice built from `(p, S)^t` on a convex monomial set reaches at most $\delta = 1/21.8$ for S_4 and $1/70.8$ for S_5. Those are 5.5× and 14.2× short of the milestone and the goal, whatever the shift strategy, the LP included.

1. Question

Is there a concrete lattice, embedding or algebraic structure that gives a decomposition oracle for Semaev's S_5 (m = 4 summands) at $\delta \approx 1/5$, or for S_4 (m = 3) at $\delta \geq 1/4$, on prime-field curves? Under §5 of `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md`, only such a structure changes the exponent. Constant-factor tuning does not count. Generic table oracles, which converge to BSGS, are out of scope.

2. Scope, inputs, and what was not done

- **Read.** `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` (all of it, including the Probe B/C tables). `RESEARCH_IC_LEADS_AND_CONNECTIONS_20261010.md` §8–§10 (from the worktree copy `worktrees/codex-n83-d18-k2048-20261027/`). The AGENTS.md pointers file. The idea-generator contract and the inventor protocol in `crypto-autoresearcher`. Its ECDLP frontier rows `KR-IC-5931ee` and `KR-IC-5a43e5`. `KN-LIT-d8e1b9`.
- **Not done: no shell was available in this session.** The available tools were file read/write, grep/glob and web search/fetch. So:
  - I could not run `git log` or `git show` on the other sessions' branches. `research/prime-fall-degree-20261027`, `research/ic_prime_residual_lp_20261026` and `lead/ic-residual-lp-20261028` are not in the local refs. Only `research/fall-degree-bounds-20261008` and `research/prime-ksum-table-20261028` are, and branch contents cannot be read without git. §6 therefore scopes by topic, not by reading their results.
  - `pandoc` and `rsvg-convert` could not be executed here. The parent session rendered `report.pdf` afterwards (§9).
  - No experiment was run and no canonical IDs were allocated.
- **Write location.** The brief asked for `/Volumes/SSD990/crypto/research/prime_s5_small_root_leads_20261028/`. This session's write hook refused that path because it is the base checkout, not this session's worktree. The files are at `/Volumes/SSD990-2/crypto/objective-ritchie-6bf995/research/prime_s5_small_root_leads_20261028/`. Copying the directory across is a plain `cp -R`.
- **The `research-visuals` skill file.** It was not on disk in either checkout (it is presumably on `origin/main`). I followed the brief's version of it: a report, an editable SVG, and a PDF.

3. The baseline, re-derived in usable units

3.1 The target in cost-exponent form

`[derived from program note §2.4]` Write $B = p^{\delta}$ and $T_{dec} = p^{\tau}$. Then cost(B) $\approx (m!/2^{m)\cdot p\cdot T_{dec}/B}\{m-1\} + m\cdot B^2$ has exponents $1 - (m-1)\delta + \tau$ and 2δ. The cost beats $\sqrt{p}$ if and only if

$$\delta < 1/4 \quad \text{and} \quad \tau < (m-1)\cdot\delta - 1/2.$$

m	polynomial	feasible δ window at τ = 0	largest admissible τ	generic oracle τ = (m−1)δ
3	S_4	none; tie only at δ = 1/4, τ = 0	0	always above
4	S_5	[1/6, 1/4)	1/4	always above
5	S_6	[1/8, 1/4)	1/2	always above
m	S_{m+1}	[1/(2(m−1)), 1/4)	(m−3)/4	always above

So an oracle need not run in polynomial time. At m = 4 it may cost up to p^{1/4}. What it must do is put its cost curve τ(δ) inside the triangle $\delta \in (1/6 + \tau/3, 1/4)$, $\tau < 3\delta - 1/2$ (Figure 1B). Every known route is listed below. Past δ = 1/24, none of them is below τ = 3δ.

- Enumerate three coordinates and root-find the fourth: τ = 3δ.
- Enumerate k coordinates and use a lattice for the rest:
  - k = 2, bivariate degree-8 lattice: valid only for δ ≤ 1/24, with τ = 2δ.
  - k = 1, trivariate: δ ≤ 1/48, with τ = δ.
  - pure lattice: δ ≤ 1/80, with τ = 0.

3.2 Finite-level lattice reach (determinant bounds)

`[derived]` In each case below, the reach at level ℓ is Σ_α k(α) divided by Σ_α (log_B of the size of x^α), taken over the lattice monomials. HG dimension factors and binomial constants are ignored, exactly as in the program note's §2.2.

- **Box (Jochemsz–May, monomials [0, dℓ]^m, d = 2^{m−1}).** $\delta_\ell = \Sigma_{i<\ell} (d\cdot i + 1)^m / (m\cdot(d\ell+1)^m\cdot d\ell/2)$, with dimension (dℓ+1)^m. This reproduces the note's 1/18, 1/10 and 0.119 at m = 2.
- **Symmetric (monomials in $e_1..e_m$, total degree ≤ Dℓ, D = 2^{m−1}).** The weights are w_i = i because e_i ≤ C(m,i)·B^i. The leading monomial is e_m^D. $\delta_\ell = \Sigma_{i<\ell} C(D\cdot i + m, m) / ((m(m+1)/2)\cdot C(D\ell + m, m + 1))$, with dimension C(Dℓ + m, m). This reproduces the note's 1/12 at m = 2, level 1.
- **The two forms have the same asymptote.** Both tend to 2/(m(m+1)·2^{m−1}) = 1/(m(m+1)2^{m−2}).
  - This is consistent with Probe C, which found equal thresholds for Rect and Sym.
  - At m = 2 the asymptote is 1/6. The note's "1/9 for the symmetric lattice" (§4.6) appears to undercount. That does not affect any decision.
  - At equal dimension the symmetric lattice is better at every finite level. At m = 2, level 3 it reaches 0.131 at dimension 28, against 0.119 at dimension 49 for the box. So it is the right vehicle for measuring m = 3 and m = 4.

m	lattice	level ℓ	dim	δ_ℓ	B at 40 bits	B at 56 bits	short-fall vs needed δ
2	box (measured HG 0.10, heuristic 0.14–0.16)	1–3	9–49	1/8.4 [measured] 0.119	48–229 [measured]	48–229 [measured]	3–5× vs 1/2
3	box	2	729	1/69.4	1.5	1.7	17× vs 1/4
3	sym	1	35	1/210	1.1	1.2	52×
3	sym	2	165	1/55.0	1.7	2.0	13.8×
3	sym	3	455	1/40.7	2.0	2.6	10.2×
3	sym	4	969	1/35.5	2.2	3.0	8.9×
3	asymptotic	∞	∞	1/24	3.2	5.0	6×
4	sym	1	495	1.26e–4	1.0	1.0	≈1600× vs 1/5
4	sym	2	4845	1/313	1.09	1.13	≈63×
4	sym	3	20475	1/184	1.16	1.23	≈37×
4	asymptotic	∞	∞	1/80	1.41	1.62	16×

The symmetric rows assume premise P2 of L4-B: the symmetrised support is the full simplex of degree 2^{m−1} with e_m^{2}{m−1}} present.

One [measured] observation matters for interpretation. At m = 2, level 1, the heuristic success at 40 bits extends to δ = 0.14, which is 2.5 times the level-1 bound of 1/18. At 56 bits it drops to 0.10–0.12. So the slack shrinks with p, which is what a finite-size effect looks like (L4-D).

3.3 Six structural facts

- **F1. The ideal is principal.** **[derived from known irreducibility]**

- For generic z = x_R, the zero set of S_{m+1}(·; z) in A^m is x^m(A_R), the image of the coset A_R = {(P_i) : ΣP_i = R} ≅ E^{m−1}.
- A_R is irreducible. The images of all sign patterns coincide because x(−P) = x(P). So the ideal of the zero set is (S_{m+1}(·; z)), given that S is squarefree.
- Hence every polynomial that vanishes on all decompositions is a multiple of S. That includes translated targets S_{m+2}(x, x_T, x_{R+T}), symmetrised forms, and resultant splittings. The curve structure contributes nothing beyond S itself.
- Irreducibility of S_n is known: Semaev 2004 `[recalled]`, and the corpus re-derives it via the Galois group (Z/2)^{m−2} (KN-FIND-a8990a, cited from KR-IC-5a43e5 `[kb]`).

- **F2. Newton polytopes are rigid.** **`[derived; Ostrowski, recalled]`**
  - Over F_p, NP(g·S) = NP(g) + NP(S). So every nonzero element of the ideal mod p has a Newton polytope containing a translate of NP(S). No sparser or lower-degree representative exists, and the JM monomial sets are forced up to shifts.
  - Modulo p^ℓ, zero divisors weaken this into a heuristic. That is why P3 asks an LP.
- **F3. x is degree-optimal among coordinates.** **`[derived]`**
  - Every degree-2 function in F_p(E) has the form M∘x∘τ_S, where M is a Möbius map and S ∈ E(F_p). Its involution is P ↦ Q − P, and Q can be halved because the group order is odd.
  - A degree-D coordinate gives a relation of degree up to D^{m−1} in each variable. With R fixed, P_1 has D^{m−1} candidates.
  - Values are identified only when the fibres are Aut(E)-orbits, so D ∈ {2, 4, 6}. Then |Fl ≈ D·B, and the per-|Fl reach 2/(m(m+1)D^{m−1}) is largest at D = 2.
- **F4. Box volume is invariant.** **`[derived, first order, heuristic]`**
  - Consider substitutions of degree 1 in new box-bounded unknowns of the same total volume. Examples are Farey (x = u/v), lattice cosets (x = u + λv), base-M digits on Solinas primes, and Möbius maps.
  - They keep the degree pattern. To first order, the JM reach measured in box volume is unchanged. The linear case is the template: aX + bY ≡ c is solvable up to XY ≈ p, exactly as x alone is up to p.
- **F5. Overdetermination costs B/p.** **`[derived]`**
  - In a decomposition search the expected number of solutions in a box of volume V is V/p. Each further independent congruence multiplies it by about B/p (or 1/p).
  - Many-sample Coppersmith gains depend on a planted consistent solution, so they do not transfer. Examples are EC-HNP with 1/(d+1) known bits (Xu–Sarkar–Wang–Hu 2022 `[literature, abstract]`) and MIHNP.
  - By F1, there are no free equations to add.
- **F6. The global-lifting wall.** **`[literature + derived]`**
  - Turning a congruence into an exact equation on small inputs means making |value| < p. That forces a relation over a number field.

- With random targets this happens only below the Coppersmith regime, or as xedni-type relations. Xedni-type relations do not beat rho (Jacobson–Koblitz–Silverman–Stein–Teske 2000 `[recalled]`, program note §4.6).

Conditional closure statement `[proposed]`. On a prime-order curve, every lattice oracle over a coordinate factor base is bounded by the shift-set LP of S_{m+1}'s Newton polytope. That bound is the JM value if P1–P3 hold. L4-B tests exactly this. The §4 closure standard of the inventor protocol is met only after L4-B's premises have been measured. Until then the closure is `unverified`.

4. Ranked idea records

No canonical IDs were allocated (no shell, and the brief forbade edits outside this directory). The local labels are L4-A to L4-D. Each record follows the `idea` schema of `agents/idea-generator.md`, abbreviated where a field does not apply.

Rank 1: L4-B, the ideal-rigidity ceiling (closure test)

```
idea:
  id: "unallocated (local label L4-B)"
  title: "Ideal-rigidity ceiling for lattice decomposition oracles on coordinate
factor bases"
  class: mechanism
  claim: >
    [proposed] On a prime-order E/F_p, every Coppersmith-type lattice oracle for
    m-decompositions over any factor base {P : t(P) in a box}, t in F_p(E), has
    asymptotic reach delta <= 1/(m(m+1)2^(m-2)) (1/24 at m=3, 1/80 at m=4),
provided
    P1 S_{m+1}(.;z) is absolutely irreducible for a 1-O(1/p) fraction of z;
    P2 NP(S_{m+1}(.;z)) = [0,2^(m-1)]^m, equivalently the symmetrised support is
the
       full degree-2^(m-1) simplex with e_m^(2^(m-1)) present;
    P3 the automated-Coppersmith LP (Meers-Nowakowski 2023; Ding et al. 2026)
over shift
       sets drawn from (S,p)^l returns the JM value for that support.
  mechanism: >
    F1 fixes the information (every polynomial vanishing on all decompositions is
a
    multiple of S). F2 fixes the support (no multiple is sparser). F3-F4 fix the
    coordinates (alternatives are box-preserving substitutions or raise the
degree).
    The only remaining freedom is the shift set, which P3 prices exactly.
  novelty_status: adaptation   # optimality-of-lattices results exist (Aono et
al.); applying them
                               # to Semaev's family with F1-F4 is not found in
the sources checked
  citations:
    - {ref: "ePrint 2012/108; ACISP 2012; AAECC 29(2) 2018 (Aono-Agrawal-Satoh-
Watanabe)", provenance: retrieved,
```

```
      claim: "bounds the reach of lattice constructions; JM/Coppersmith-May
constructions asymptotically optimal in their framework",
      verified_by: "search-result abstract, 2026-10-28"}
    - {ref: "ePrint 2023/1409 (Meers-Nowakowski, Asiacrypt 2023)", provenance:
retrieved,
      claim: "combinatorial, automatable restatement of Coppersmith",
verified_by: "search-result abstract"}
    - {ref: "ePrint 2026/1027 (Ding-Dai-Wu-Zhang-Zhang)", provenance: retrieved,
      claim: "asymptotic bounds of automated Coppersmith computed by linear
programming", verified_by: "abstract page fetched"}
    - {ref: "Semaev, ePrint 2004/031", provenance: recalled, claim: "S_n
symmetric, irreducible, degree 2^(n-2) per variable"}
    - {ref: "KR-IC-5a43e5 / KN-FIND-a8990a", provenance: kb, claim: "Galois group
(Z/2)^(m-2); re-derived irreducibility"}
  proof_search_map:
    bottleneck: "reach of the shift set for a fixed polynomial support;
everything else is fixed by F1-F4"
    baseline_embedding:
      parameter_slice: "m=2 reproduces Probe B/C: box levels 1/18, 1/10, 0.119;
sym level 1 = 1/12; both asymptotes 1/6"
      reproduction_check: "closed forms in README section 3.2 (symbolic)"
    observation_collision:
      observable: "planted-root reach in the Probe B protocol"
      distinct_preimage_search: "random polynomial with S4's exact support and
degree pattern; equal reach => Semaev structure not load-bearing"
    constructive_transforms:
      - {transform: representation_reduction, proposed_object: "support polytope
+ LP", predicted_gain: "none (predicts JM)"}
    quantifier_order: "for all prime-order E, all coordinate t, all shift sets in
(S,p)^l: reach <= LP(NP(S)); P1-P2 checked for sampled z"
    method_ceiling:
      strongest_certifiable_claim: "ceiling for lattice oracles only, conditional
on Coppersmith's heuristic and P1-P3"
      nearby_object_control: "even-order curve with rational 2-torsion: the
closure must predict only a constant change (K3)"
    proof_obligations:
      - {claim: "P1 irreducibility of specialisations", responsibility:
correctness}
      - {claim: "P2 support = full box/simplex", responsibility: size}
      - {claim: "P3 LP optimum = JM", responsibility: strictness}
      - {claim: "prime order (no rational torsion) is required", responsibility:
scope}
  predictions:
    - {metric: "nonzero vertex coefficients of S4(.;z), 40-bit Probe-B curve, 20
z", direction: higher, minimum_effect: "8/8 vertices for >= 19/20 z"}
    - {metric: "LP optimum / JM value (m=3, m=4)", direction: different,
minimum_effect: "within 1% of 1"}
    - {metric: "max B with >= 4/8 success: S4 vs null polynomial at equal dim",
direction: different, minimum_effect: "equal within one B step"}
```

```
  minimal_test:
    design: >
      Step 1 (symbolic, minutes): build S4(x1,x2,x3,z) by resultant for the 40-
bit Probe-B curve
      at 20 random z. Record the 8 vertex coefficients of [0,4]^3, the
symmetrised support and the
      e3^4 coefficient. Test absolute irreducibility by restricting to 50 random
lines and factoring
      over F_p and F_{p^2}. Repeat Step 1 for the symmetrised S5 support.
      Step 2 (LP, minutes): feed both supports to the Ding et al. LP and compare
to 1/24 and 1/80.
      Step 3 (Probe B/C framework, hours): only if the LP shift set differs from
JM, run it at m=3
      on the 40-bit curve with Probe B's planted-root protocol, B in {2,3,4,6,8},
8 trials per cell,
      beside Sym{l} of equal dimension; then m=4 at level 1. Otherwise go
straight to L4-A.
    controls: ["null polynomial with identical support", "Sym{l} of equal
dimension"]
    required_metrics: ["vertex count", "irreducible-restriction fraction", "LP
optimum", "max B with >= 4/8 success", "ms per LLL"]
  falsification_conditions:
    - "Closure falsified if a vertex coefficient vanishes identically in z (P2),
S4(.;z) is reducible for > 1% of z (P1), the LP optimum exceeds JM by > 5% (P3),
or an LP-optimal lattice reaches >= 2 B-steps beyond Sym at equal dimension in
Step 3."
    - "Any such failure is the one lattice lead left open here. Its best case is
a constant factor in delta unless the support shrinks by a factor that grows with
m."
  confounders: ["LLL vs BKZ quality at dim > 500", "finite-size slack at 40 bits
(B <= 3 makes the Probe test coarse)"]
  interpretation_limits: ["lattice oracles only", "prime-order curves only",
"says nothing about Groebner/fall-degree or non-coordinate factor bases"]
  heuristic_assumptions:
    - id: H1
      statement: "Coppersmith heuristic: the first m reduced polynomials are
algebraically independent at the root"
      rigorous_support: "Howgrave-Graham bound (rigorous for vanishing over Z);
independence is the heuristic"
      validation_plan: "Probe B's '>= 2 vanishing rows' and resultant non-
vanishing rate, 8-64 trials per cell; compare to the HG-holds fraction"
  target_complexity:
    time_exponent: "n/a (closure test)"
    memory_exponent: "n/a"
    best_known: "Pollard rho sqrt(pi n/4); preprocessing rho S*T^2 ~ n"
    hidden_overhead: "none claimed"
    tradeoff_note: "n/a"
  prior_art:
    rows_checked: [KR-IC-5931ee, KR-IC-5a43e5, "program note section 2.2 / 4.6"]
```

```
    nearest: "Aono et al. optimality (general); program note Probe C (empirical,
m=2)"
    relation: adaptation
    delta: "applies the optimality question to S_{m+1} with F1-F4, so the ceiling
covers every coordinate factor base, not one lattice"
  estimated_cost: {implementation: low, compute: low}
  recommended_priority: high
```

Rank 2: L4-A, the finite-level S_4/S_5 baseline (control)

```
idea:
  id: "unallocated (local label L4-A)"
  title: "Measured S4 and S5 small-root reach for the best available (symmetric)
lattices, normalised by enumeration cost"
  class: control
  claim: >
    [proposed] At m=3 and m=4 the measured reach of symmetric JM lattices matches
the section 3.2
    determinant bounds, within the m=2 heuristic slack (at most 2.5x at level 1,
shrinking with p).
    In particular, at 40 bits no lattice of dimension <= 5e3 reaches B >= 4 at
m=3 or B >= 3 at m=4
    with >= 4/8 success, and at every cell the lattice costs more per call than
the B^(m-1)
    enumeration it would replace.
  mechanism: "JM determinant bound plus a finite-size slack that decays with p
(section 3.2)"
  novelty_status: known   # a measurement of known lattices; the program has not
run m >= 3
  citations:
    - {ref: "RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md section 3.2-3.3",
provenance: internal, claim: "m=2 measured reach; protocol"}
    - {ref: "Jochemsz-May, Asiacrypt 2006", provenance: recalled, claim: "basic
and extended strategies"}
  predictions:
    - {metric: "max B with >= 4/8 success, m=3, Sym l=2..4", direction: lower,
minimum_effect: "B in [2,3] at 40 bits; [2,5] at 56 bits"}
    - {metric: "max B with >= 4/8 success, m=4, Sym l=1..2", direction: lower,
minimum_effect: "B <= 2 at 40 and 56 bits"}
    - {metric: "false-positive rate on unplanted targets", direction: lower,
minimum_effect: "0/8 per cell"}
  minimal_test:
    design: >
      Extend examples/prime_ecdlp_breakthrough_probe.rs with S4 and S5 (by
resultants) and their
      symmetrised forms. Curves at 32, 40, 48 and 56 bits. Planted roots with B
in {1,2,3,4,6,8,12,16};
      8 planted and 8 unplanted targets per cell. Lattices Sym{1..4} at m=3 and
Sym{1,2} at m=4
      (dim 495, 4845), plus box level 2 at m=3 (dim 729) as a cross-check. Record
```

```
HG-holds,
      '>= 2 vanishing rows', ms per reduction, and enumeration cost B^(m-1) for
the same cell.
    controls: ["unplanted targets (decision behaviour)", "null polynomial with
S4's support (from L4-B)", "m=2 Probe B rows re-run as a regression"]
    required_metrics: ["delta reach per (m, l, bits)", "slack = heuristic reach /
delta_l", "ms per LLL / B^(m-1)"]
  falsification_conditions:
    - "Heuristic success >= 4/8 at delta >= 1/24 at m=3 (B >= 4 at 40 bits, B >=
6 at 56 bits) on >= 2 sizes, with reach not decreasing in p: that is structure-
specific slack beyond the asymptotic JM value, and it contradicts L4-B."
    - "Success at B >= 3 for m=4 at level <= 2: contradicts the section 3.2 table
and reopens the S5 lattice question."
  confounders: ["LLL non-determinism across implementations", "dimension factors
in HG at dim > 1000", "tiny B makes brute force trivially cheaper"]
  interpretation_limits: ["a 40-bit curve resolves delta only in steps of
log2(B)/40; the milestone is B = 1024 at 40 bits, so any lead is unambiguous but
small reaches are coarse"]
  heuristic_assumptions:
    - id: H1
      statement: "as in L4-B"
      rigorous_support: "Howgrave-Graham"
      validation_plan: "fraction of '>= 2 vanishing' cells where the resultant
actually recovers the root"
  target_complexity:
    time_exponent: "n/a (control)"
    memory_exponent: "n/a"
    best_known: "enumeration tau=(m-1)delta; rho p^(1/2)"
    hidden_overhead: "lattice cost poly(dim) with dim up to 4845"
    tradeoff_note: "n/a"
  prior_art: {rows_checked: [KR-IC-5931ee], nearest: "Probe B/C (m=2)", relation:
extension, delta: "first m=3/4 rows; corrects the quoted '1/80 measured'"}
  estimated_cost: {implementation: medium, compute: "low for m=3; medium-high for
m=4 level 2 (dim 4845 reductions)"}
  recommended_priority: high
```

Rank 3: L4-D, the probabilistic regime (success-probability decay past the rigorous bound)

```
idea:
  id: "unallocated (local label L4-D)"
  title: "Can the section 2.4 allowance T_dec <= p^((m-3)/4) be spent on lattice
luck? Measure q(delta,p) past the bound"
  class: measurement
  claim: >
    [proposed] Past the rigorous bound, lattice success on decomposable targets
decays as
    q = p^(-gamma(delta)), with gamma growing at least linearly in (delta -
delta_l) times the
    dimension. Then the probabilistic oracle exponent tau = gamma(delta) exceeds
```

```
enumeration's
    (m-1)delta for every delta above the asymptotic JM value, and the allowance
cannot be spent on
    luck. Competing explanations: (i) finite-size slack, where q at fixed delta
tends to 0 as p
    grows (predicted); (ii) structural slack, where gamma saturates below
(m-1)delta (this would
    open a probabilistic-oracle lane).
  mechanism: "beyond the Minkowski/HG regime the target vector is no shorter than
typical lattice vectors; success needs a short-vector coincidence whose
probability falls with dim*(delta-delta_l)*log p"
  novelty_status: unverified
  citations:
    - {ref: "program note section 3.2 Probe B table", provenance: internal,
       claim: "m=2 level 1, delta=0.12: 7/7 (40-bit) vs 3/8 (56-bit); level 3,
delta=0.14: 7/8 vs 3/8; delta=0.16: 5/8 vs 0/8 [measured]"}
  predictions:
    - {metric: "fitted gamma(delta) at delta = 1.2*delta_inf (m=2,3)", direction:
higher, minimum_effect: "gamma > (m-1)*delta"}
    - {metric: "q at fixed delta vs bits (32..64)", direction: lower,
minimum_effect: "monotone decrease, slope significant at 2 sigma"}
  minimal_test:
    design: >
      Probe B/C framework. m=2 at levels 1-3 and m=3 Sym{1,2}; curves at 32, 40,
48, 56 and 64 bits;
      delta grid from delta_rig to 2*delta_inf; 64 planted trials per cell. For
each fixed delta,
      fit log2 q against log2 p to get gamma(delta). For m=3, compare with
(m-1)delta = 2 delta.
      Re-randomise per trial by a fresh target, not by re-running LLL on the same
basis
      (LLL is deterministic).
    controls: ["unplanted targets (false-success rate)", "null polynomial with
the same support"]
    required_metrics: ["q(delta, bits) with Wilson intervals", "gamma(delta) with
standard error", "ms per call"]
  falsification_conditions:
    - "Prediction falsified if some delta > delta_inf has a fitted gamma(delta) <
(m-1)delta on >= 3 sizes with non-overlapping intervals. That would be the first
evidence of a probabilistic lattice oracle below enumeration in exponent."
  confounders: ["discrete B steps at small bit sizes", "basis-reduction quality
(LLL vs BKZ-20) changes q but should not change gamma's sign"]
  interpretation_limits: ["m=2 and m=3 only; extrapolating to m=4 assumes the
same slope law"]
  heuristic_assumptions:
    - id: H2
      statement: "q(delta,p) = p^(-gamma(delta)) * poly(log p) for delta in
(delta_l, 2 delta_inf)"
      rigorous_support: "Gaussian-heuristic short-vector counts in random
```

```
lattices of fixed determinant (imitated, not proved for these lattices)"
      validation_plan: "five bit sizes, 64 trials per cell; check log-linearity
(residuals) and that the 64-bit point lies within the 32-56-bit fit's prediction
interval"
  target_complexity:
    time_exponent: "measured tau(delta) = gamma(delta); success region requires
tau < 3 delta - 1/2 at m=4"
    memory_exponent: "poly"
    best_known: "enumeration tau = (m-1)delta"
    hidden_overhead: "poly(dim) per call"
    tradeoff_note: "none"
  prior_art: {rows_checked: [KR-IC-5931ee], nearest: "Probe B", relation:
extension, delta: "turns the threshold into a cost exponent, which is what
section 2.4 actually requires"}
  estimated_cost: {implementation: low, compute: low}
  recommended_priority: medium
```

Rank 4: L4-C, structure in the gap (Poisson and clustering test of small roots)

```
idea:
  id: "unallocated (local label L4-C)"
  title: "Are small S4 decompositions between delta = 1/24 and 1/4
distinguishable from a null model?"
  class: measurement
  claim: >
    [proposed] For delta in (1/24, 1/4), the small 3-decompositions of random
targets on prime-order
    curves cannot be told apart from null models: counts are Poisson with mean
lambda ~ c*B^3/p,
    targets R and R+T share summands no more often than chance, and
decompositions do not
    concentrate on low-height integer subvarieties. Alternative: over-dispersion
or excess
    coincidences, meaning algebraic structure in the gap that a non-lattice
(determinant-method-type)
    oracle could target.
  mechanism: "between the rigidity regime (where small points lie on few Z-
curves; Chang et al.) and equidistribution, nothing known forces structure; F1
says the only algebraic input is S itself"
  novelty_status: unverified
  citations:
    - {ref: "arXiv:1111.1543 (Chang, Cilleruelo, Garaev, Hernandez, Shparlinski,
Zumalacarregui)", provenance: retrieved,
       claim: "upper bounds for points of modular curves in small boxes; rigidity
at small box sizes", verified_by: "search-result abstract"}
    - {ref: "KN-LIT-d8e1b9 (Yokoyama-Yasuda-Takahashi-Kogure, JMC 14, 2020)",
provenance: kb,
       claim: "naive Semaev IC over F_p no better than generic, under statistical
assumptions on summation polynomials"}
  predictions:
```

```
    - {metric: "dispersion index Var/mean of decompositions per target",
direction: different, minimum_effect: "within [0.95, 1.05] (null)"}
    - {metric: "R vs R+T shared-summand rate over null 1", direction: different,
minimum_effect: "within 3 sigma"}
  minimal_test:
    design: >
      Toy prime-order curves at p ~ 2^26, 2^28, 2^30: one with a = -3 and random
b, one with a = 0 and
      small b (the secp256k1 shape). B = floor(p^delta), delta in {1/24, 1/12,
1/8, 1/6, 1/5, 1/4};
      1e5 random targets per cell. Exact enumeration over pairs (P1, P2) in F^2,
checking
      x(R - P1 - P2) <= B: B^2 point operations per target, at most 3.3e4 at p =
2^30, delta = 1/4,
      so minutes per cell on the native u64 engine. This is the Probe B/C
protocol run exhaustively at
      toy size instead of by lattice. It gives the denominator any m=3 oracle at
40 bits must beat:
      the expected number of decompositions per target.
    controls: ["null 1: same group, factor base replaced by a uniformly random
subset of E(F_p) of the same size",
               "null 2: random polynomial with S4's support in place of S4 (keeps
shape, removes the group)"]
    required_metrics: ["dispersion index with 95% CI", "pair-coincidence rate vs
nulls", "rank of degree-<=2 monomial matrix on multi-decomposition targets vs
nulls"]
  falsification_conditions:
    - "Null claim falsified if the dispersion index CI excludes 1 at >= 2 sizes
for the same delta, or the coincidence excess over null 1 exceeds 3 sigma. That
is a positive lead, to be reported with the delta and curve shape."
  confounders: ["the a=0 curve's order-6 automorphism produces orbit-correlated
decompositions (expected; count orbits, not points)", "twist x-values (x of
points on the quadratic twist) must be excluded from F"]
  interpretation_limits: ["toy p only; transfer to 40-256 bits assumes
stationarity in delta, which the three sizes check"]
  heuristic_assumptions:
    - id: H3
      statement: "N_R (number of small 3-decompositions of a random R) ~
Poisson(lambda(B,p))"
      rigorous_support: "Weil/Deligne equidistribution of points on the relation
variety in large boxes (imitated below p^(1/2))"
      validation_plan: "1e5 targets x 6 deltas x 3 sizes; empirical vs Poisson
pmf, chi-square, and tail check P(N >= 3)"
  target_complexity:
    time_exponent: "n/a (measurement)"
    memory_exponent: "n/a"
    best_known: "n/a"
    hidden_overhead: "none"
    tradeoff_note: "n/a"
```

```
  prior_art: {rows_checked: [KR-IC-5931ee], nearest: "Yokoyama et al. statistical
assumptions (unverified at these deltas)", relation: extension,
              delta: "measures the assumption in the exact regime between lattice
reach and the milestone; also validates the Poisson yield used by section 2.4"}
  estimated_cost: {implementation: low, compute: low}
  recommended_priority: medium
```

Ranking rationale and what to test first

- **L4-B goes first.** Its Step 1 is a symbolic computation on one 40-bit curve and takes minutes. Step 2 is an LP of seconds to minutes. The outcome decides whether *any* future "new lattice" proposal can move δ's exponent:
  - if P1–P3 hold, every lattice variant is closed for coordinate factor bases on prime-order curves (§3.3);
  - if one fails, the failure names the only lattice lead left.

  It is also the cheapest valid discriminator in this note, because it needs no lattice reduction. L4-A and L4-D need the support facts P2 that L4-B produces before their symmetric-lattice dimensions mean anything.
- **L4-A fills in the measured rows the brief assumed already exist.** Information is moderate, since the predictions are fairly confident. Cost is medium for m = 4.
- **L4-D asks the right §2.4 question: cost exponent, not threshold.** It is cheap. A positive result would matter, but its prior is low.
- **L4-C is the only test aimed at non-lattice structure.** It is very cheap. A null result strengthens the closure. A positive result would be the only new lead.

5. Killed on paper

Every entry is `[derived]` unless marked otherwise. "Reach" means the JM asymptotic δ in factor-base-size units.

#	idea examined	mechanism hoped for	why it fails	effect on δ
K1	Möbius-transformed or translated x-intervals; any other degree-2 coordinate	a better-conditioned box	F3: every degree-2 F_p-function is $M \circ x \circ \tau_S$. Homogenising keeps degree d per variable and the same box	none (≡ JM)
K2	Farey ($x = u/v$), GLV/lattice-coset ($x = u + \lambda v$), SNFS base-M digit bases on Solinas primes ($x = \Sigma a_j M^j$)	"approximate subfield"; Weil restriction over Z	F4: degree-1 substitution, same box volume, same first-order reach. Exact splitting into several equations over Z needs $\lvert S(\ldots)\rvert < M$, i.e. small x_R and small b: that is F6 (xedni)	none
K3	Torsion symmetrisation (FHJRV), including the "D_4-symmetrised S_5 probe"	lower-degree invariants	needs $E(F_p)[2]$ or $[4] \neq 0$, which prime-order curves lack. On even-order curves, quotienting by $\langle T \rangle$ maps the problem onto $E/\langle T \rangle$ with $\lvert F\rvert \times \lvert T\rvert$ at fixed box: a constant	constant
K4	Interval membership polynomial, [illegible] falling-factorial; Bostan–Schost products	evaluate "∃ root in the box" faster than enumeration	the m-dimensional box product costs $B^{m^2/(m+1)}$ (balance $s^{m+1} = B^d$), which is above enumeration's	$\tau = 3.2\delta > 3\delta$ at $m = 4$
K5	[illegible]	[illegible]	[illegible]	≈ 1.28 at (m = 2)

Also checked and already closed in the program note, not re-proposed: the symmetric (e_1, e_2) rewrite of S_3 (§3.2 above shows its asymptote is 1/6, not 1/9, which changes nothing); multiplicative-subgroup bases; global lifting / xedni; embedding into an extension; lattice-assisted MITM; low-degree-map factor bases.

6. Relation to the other active sessions

- **Prime-field fall degree / Gröbner** (`research/prime-fall-degree-20261027`, `research/fall-degree-bounds-20261008`, `codex/fall-degree-*`).
  - Nothing here proposes a Gröbner run.
  - K12 hands the Boolean-variable encoding to them.
  - If their solving degrees for S_4/S_5 with interval constraints come out sub-enumerative, that lies outside F1–F4, which are lattice-specific, and would supersede this note's closure for that solver class.
  - I could not read their branches (§2).
- **Residual large-prime IC** (`research/ic_prime_residual_lp_20261026`, `lead/ic-residual-lp-20261028`). Large-prime variants change constants in the relation phase. They do not change the §3.1 exponent condition, because the oracle's τ(δ) is unaffected. Not duplicated.
- **k-sum table** (`research/prime-ksum-table-20261028`). Generic, out of scope by the brief. Figure 1B's black line is its exponent.

7. Honest accounting (inventor protocol §5)

```
honest_accounting:
  objects:
    - "tracked object: the decomposition tuple (x_1..x_m) as a small point on
V(S_{m+1}(.;z)); operation set: none beyond the ideal (S)"
    - "representations examined: K1-K16 (coordinate changes, divisor/line/orbit/
bit bases, lifts, covers)"
  lossy_projection_test: >
    No new tracked object is proposed. Of the representations examined, K1, K2
and K11 are lossless
    changes of coordinates (they fail the test). K6-K9 are lossy but raise the
degree. Index calculus
    stays Class I partial-action (IDEA-20260806-c5d183). None of K1-K16 escapes
KN-OPEN-020: all are
    explicit low-degree descriptions.
  depth_of_verified_structure: "paper-level (derived) only; nothing measured this
session"
  dominated_by: "n/a (no result claimed). Rows checked: Pollard rho sqrt(pi n/4);
preprocessing rho (S*T^2 ~ n, Probe A); generic IC (m-1)-enumeration; JM
asymptotic lattice; McGuire-Mueller O(p log^2 p) (ePrint 2017/1262); naive Semaev
IC (KR-IC-5931ee)"
  sota_delta: >
    No attack. Corrected baseline: at feasible dimension (<= 5e3) the S5 reach is
about 1/313
```

```
    against 1/5 needed (about 63x), and the S4 reach is 1/35.5 against 1/4
(8.9x). Asymptotic
    gaps are 16x and 6x.
  closures:
    - "lattice oracles on coordinate factor bases, prime order: F1+F2+F3+F4 ->
shift-set LP; status unverified until L4-B measures P1-P3"
    - "many-sample / overdetermined lattice imports: F5 (no planted solution, no
free equations)"
    - "exact-vanishing tricks (small coefficients, digit bases, quadratic
orders): F6 (global lifting, xedni)"
    - "K1-K16 individually, each with its mechanism in section 5"
  open_directions:
    - "a non-lattice small-root algorithm for one structured congruence at box
volume p^0.7-0.8, which would need a named external ingredient; none found
(searches below)"
    - "factor bases whose membership is implicit, high-degree or target-dependent
(the KN-OPEN-020 escape), with a polynomial-time membership test and no
coordinate-box description"
    - "transfer (program note section 4.3) and Groebner/fall degree (other
sessions)"
    - "if L4-C shows over-dispersion: a determinant-method oracle aimed at the
observed structure"
```

7a. L4-B step 1, run afterwards (measured, 2026-10-28)

The parent session ran L4-B's first step with PARI/GP (`l4b_step1.gp`, output `l4b_step1.out`). Setup: the frontier campaign's 40-bit `a = -3` curve `icv1-fp40-tm924137-0ecaaeb9` ($p = 2^{40} - 213$, $b = 56$); 20 seeded random z; $S_4 = \mathrm{Res}_X(S_3(x_1,x_2,X), S_3(x_3,z,X))$. It takes under a second.

- **[measured] P2 holds on every sample.** For all 20 values of z, all 8 vertex coefficients of the box $[0,4]^3$ are nonzero, so the Newton polytope of $S_4(\cdot; z)$ is the full box. No sparser support exists for a lattice to exploit at $m = 3$.
- **[measured] Irreducible over F_p on every sample.** For all 20 values of z, a restriction to a random line has full degree 12 and is irreducible over F_p.
- **[not tested] Absolute irreducibility.** Line restrictions cannot decide it. A degree-12 restriction that is irreducible over F_p splits over F_{p^k} into $\gcd(k, 12)$ factors whatever S_4 does. A first version of the script that "tested" F_{p^2} was wrong for this reason and was corrected. Absolute irreducibility of summation polynomials is [recalled] a known result (Diem); it was not re-read here.
- **P3 not run.** The automated-Coppersmith linear program of Ding et al. (ePrint 2026/1027) has no implementation in this repository. §7c bounds it anyway for x-variable lattices.
- **[measured] $m = 4$ (S_5), the arity where beating rho is possible.** `l4b_step1_s5.gp` (`l4b_step1_s5.out`): $S_5 = \mathrm{Res}_X(S_3(x_1,x_2,X), S_4(X,x_3,x_4))$ on the same curve, 20 seeded z.

- All 16 vertex coefficients of $[0,8]^4$ are nonzero for all 20 values, so P2 holds.
- A restriction to a random line has full degree 32 and is irreducible over F_p for all 20 values.

Reading. The failure of P2 was the only lattice lead L4-B could open at $m = 3$ or $m = 4$, and P2 holds at both. P1 holds over F_p. Only P3, the linear program, is left open. Under L4-B's claim, every lattice oracle on a coordinate factor base is then capped at $1/(m(m+1)2^{m-2})$: $1/24$ at $m = 3$ and $1/80$ at $m = 4$, against the $1/4$ milestone and the $1/5$ goal. **Lead 4 is closed for lattice methods on coordinate factor bases**, unless P3 returns a better shift set than Jochemsz–May. What remains is the non-lattice space of §7, where this note has no concrete candidate.

7b. Lead-4 code and the follow-up probes (measured, 2026-10-28)

Five pieces of native Rust, all on main's crate and with no off-main dependency:

- `src/cryptanalysis/semaev_prime_poly.rs`: dense $S_3 \ldots S_6$ over F_p by the GP recursion, plus the structure checks;
- `examples/semaev_prime_structure.rs`: L4-B step 1, natively;
- `examples/semaev_gap_census.rs`: L4-C;
- `examples/semaev_lattice_reach.rs`: L4-A;
- `examples/semaev_shift_search.rs`: the P3 fallback.

Each probe has a hashed `PROTOCOL.md`, registered before its measured run (`REGISTRATION.sha256`), and the hashes of the binary and source it ran (`BINARY.sha256`). Raw outputs are under `*/data/`. Derived tables and figures come from `*/derive.mjs`, which reads only those outputs.

After the runs, CI's clippy (`-D warnings`) required five lint-only edits to `semaev_prime_poly.rs` and two to `examples/semaev_gap_census.rs`: `is_multiple_of`, `div_ceil`, `repeat_n`, and $2 \cdot d < \text{len}$ for $2 \cdot d \le \text{len} - 1$. Each edit computes the same values, and the 6 unit tests still pass. The source hashes in the probes' `BINARY.sha256` refer to the sources as run, at commit f91405e.

7b.1 The GP checks, ported (L4-B step 1, native)

`semaev_prime_poly` builds `S_{m+1}(x_1…x_m; z)` as `Res_X(S_3(x_1,x_2,X), S_m(X,x_3,…; z))`, with formal-degree Sylvester resultants and tensor interpolation.

- **[measured] The values match PARI/GP exactly.** The unit test `values_match_pari` checks S_3, S_4 and S_5 at three points each on the 40-bit curve, against `polresultant` output.
- **[measured] S_4 and S_5 vanish on real sums.** Tested at $x(P_1+P_2+P_3)$ and $x(P_1+\ldots+P_4)$.
- **Irreducibility over F_p.** It is decided by intersecting factor-degree patterns of line restrictions (distinct-degree factorisation). A factor of total degree a forces a to be a subset sum on every full-degree line. This replaces the GP script's "one irreducible line" test. That

test needs about `n` lines for a degree-`n` restriction, and in a first native run it missed for S_5 (degree 32) within 60 lines.

- **[measured, `l4b/native_step1.jsonl`]** On the same curve and the same count (20 seeded `z` per arity): S_4 and S_5 have the full box (8/8 and 16/16 vertices nonzero on 20/20 samples). They are irreducible over `F_p` on 20/20, settled within at most 8 lines for S_4 and 6 for S_5.
  - New: the polynomials are **completely dense**. All 125 coefficients of $S_4(\cdot; z)$ on $[0,4]^3$ and all 6561 of $S_5(\cdot; z)$ on $[0,8]^4$ are nonzero on every sample.
  - It takes 0.1 s (S_4) and 14 s (S_5) for all 20 values of `z`.
- **[derived] Simplex vertices.** For the symmetrised form `g(e)`, the vertex coefficient of `e_i^D` equals the box coefficient of `Π_{j≤i} x_j^d`. That is the only `x`-monomial of that multidegree among `e^β` with `|β| ≤ D`. So box vertices nonzero implies simplex vertices nonzero, which is the premise §7b.4 needs.

7b.2 L4-C: is there structure in the gap? (census, `l4c/`)

The protocol changed the README record in three ways before any data:

- an exact census over all `r − 1` targets replaces 10^5 samples;
- the counting unit is fixed: signed multisets summing to `R`;
- null 2 is dropped, because its expected root count is below 10^{-2} per cell.

A 20-bit smoke run, before registration, showed that a per-target Poisson null is wrong for a finite factor base. Its triple sums collide only through short additive relations, which come in clumps. So the registered comparison is against null 1 (five random factor bases of the same size), with the primary statistic `triple_unshared`: 6-term relations among the factor base.

- **Panel.** Six prime-order curves (`a = −3` and `a = 0` at 26, 28 and 30 bits), with $\delta \in \{1/24, 1/12, 1/8, 1/6, 1/5, 1/4, 0.3\}$. That is 41 cells (one skipped for memory, as registered), 246 censuses.
- **[measured] Null result.** In all 41 cells, the small-x values of `triple_unshared`, `pair_collisions` and `ge2` lie inside the registered band (null-1 range ± 3σ). No (shape, δ) is outside on the same side at two sizes. The falsifier did not fire.
- **[measured] δ ≤ 1/5 is uninformative.** Every count is zero, for the small-x and the null bases alike, as the protocol's limits anticipated. The informative cells are $\delta = 1/4$ and 0.3 (Figure 2).
- **Reading.** Up to the toy sizes tested, small-abscissa factor bases have no more short additive relations than random factor bases of the same size. There is no gap structure for a non-lattice oracle to target. This supports the Poisson-yield assumption in §2.4 and does not reopen lead 4.

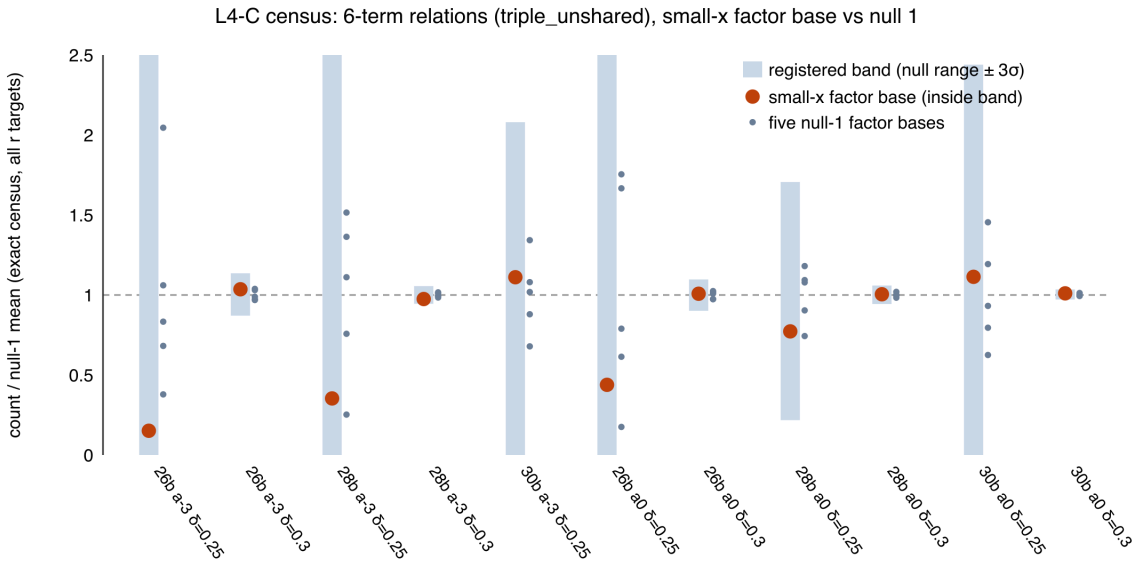


Figure 2: L4-C: 6-term relations among the factor base, small-x vs five null-1 bases (exact census over all targets; band = registered null range ± 3σ). Source: l4c/analysis/l4c.svg.

7b.3 L4-A: measured S_4 lattice reach (`l4a/`)

Symmetric JM lattices for S_4:

- $g(e_1,e_2,e_3)$ is found by a 35-unknown linear solve, checked on 6 extra points;
- the lattice is triangular in (β_3, β), with g monic in e_3^4;
- reduction uses main's `lattice::lll_reduce`.

A timing pilot fixed the scope before registration:

- dim 35 takes about 20 ms;
- dim 165 takes 116 s, about 200 s under load;
- no dim-455 reduction finished in 5 h 10 min (`l4a/pilot/PILOT.md`).

So **$\ell \geq 3$ at m = 3, and all of m = 4, stay `[derived]` only.**

Measured cells:

- **$\ell = 1$ (dim 35):** 32, 40 and 48 bits; B ∈ {1, 2, 3, 4, 6, 8, 12, 16}; 8 planted and 8 unplanted targets per cell.
- **$\ell = 2$ (dim 165):** 32 and 40 bits; B ∈ {1, 2, 3, 4, 6}; 4 planted and 4 unplanted per cell.

[measured] v1: the registered falsifier fired, on a planting defect. At $\ell = 2$ on 40 bits, B = 4 succeeded on 3 of 4 targets and B = 3 on 2 of 4, while the §3.2 bound is B ≈ 1.7.

- PARI/GP shows that **all 8 distinct successful planted targets** in v1, across every cell, satisfy $S_4(x; z) = 0$ over $\mathbb{Z}$ (`l4a/succ.gp`, `l4a/succ.out`).
- They have $z \in \{1, 2, 3, 25\}$, and most are trivial $P - P + Q = Q$ decompositions.

- Cause: small x's repeat, and the planting rule accepted any root. Any lattice that contains S_4 finds such roots automatically, which is F6 global lifting, not modular reach.
- The v1 data, protocol and fired falsifier are kept (`l4a/data-v1/`, Figure 3b). `AMENDMENT-1.md`, hashed before the rerun, redraws planted roots that vanish modulo the independent prime $2^{61} - 1$ and changes nothing else.

[measured] v2 (amendment 1): no planted success in any cell.

bits	ℓ	dim	planted success, all B	unplanted	measured reach	§3.2 bound B_ℓ	median LLL time
32	1	35	0/64	0/64	$B < 2$	1.11	8–19 ms
40	1	35	0/56 ($B = 1$: no non-global root)	0/64	$B < 2$	1.14	9–22 ms
48	1	35	0/64	0/64	$B < 2$	1.17	10–24 ms
32	2	165	0/20	0/20	$B < 2$	1.50	46–131 s
40	2	165	0/16 ($B = 1$: no non-global root)	0/20	$B < 2$	1.66	66–178 s

Every reduction costs far more than the $(B + 1)^2$ enumeration it would replace: 46–178 s against at most 49 candidates at $\ell = 2$.

Reading. The measured m = 3 reach agrees with the §3.2 determinant bounds. They predict $B < 2$ at both levels and all three sizes, and v2 finds no success at $B \geq 2$. There is no slack beyond the bound, unlike the up-to-2.5× slack Probe B saw at m = 2. No falsifier fired in v2. The m = 3 rows at $\ell = 1$ and 2 are now `[measured]`; $\ell \geq 3$ and all of m = 4 remain `[derived]`. The v1 episode is a lesson for any future planted-root test: exclude global relations, or a lattice "succeeds" on them by construction.

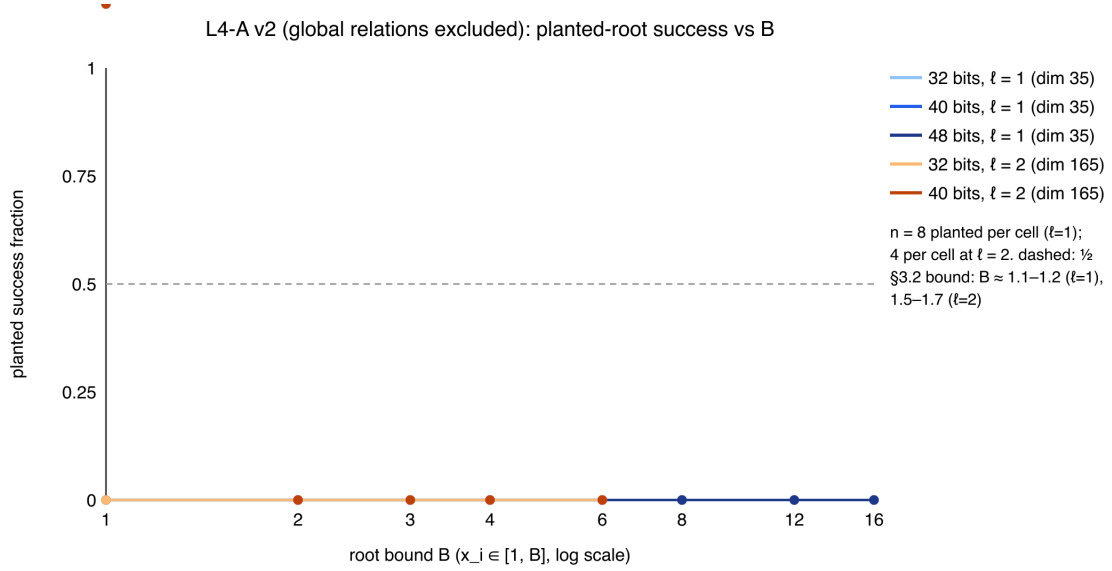


Figure 3: L4-A v2: planted-root success of symmetric JM lattices for S_4 against the root bound B, with global relations excluded (zero in every cell). Source: l4a/analysis/l4a.svg.

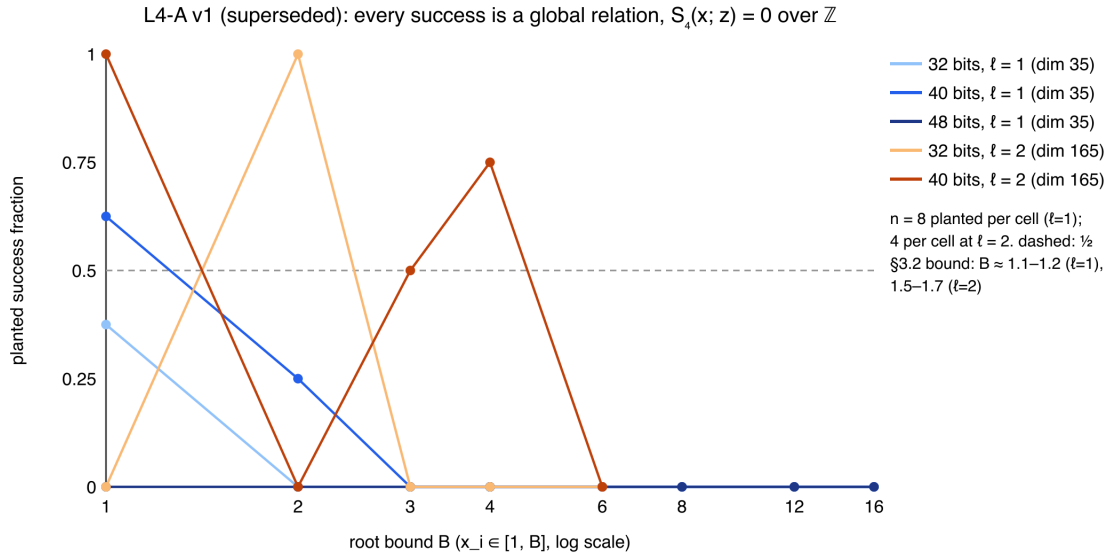


Figure 4: L4-A v1 (superseded): the apparent successes, every one a global relation $S_4(x; z) = 0$ over $\mathbb{Z}$. Source: l4a/analysis/l4a-v1.svg.

7b.4 P3 fallback: our search over triangular shift lattices (`p3/`)

What this is not. P3 names the LP of Ding et al. (ePrint 2026/1027). Its PDF is behind a Cloudflare browser challenge this session does not pass, so that LP is **not implemented here**.

What was run instead. An exact count over an explicit family of triangular shift lattices:

- every leading vertex of the Newton polytope;
- monomial sets made of rectangles with extended caps, cut by `|β| ≤ s` or by one `{0,1,2}` half-space;
- `k(β)` maximal subject to the shifts staying in the set.

The score is the determinant bound `δ = Σk / Σ⟨w,β⟩`, without dimension factors. The code reproduces every §3.2 JM value exactly (`jm_set_delta = jm_formula` in all rows). That covers the m = 4 symmetric values: $1.26 \cdot 10^{-4}$, 1/313 and 1/184.

shape	m	best δ found (dim)	best δ / δ_JM at equal dimension	asymptote	exceeded
simplex (e)	3	0.0326 (3642)	1.078 at dim 56	1/24 = 0.0417	no
box (x)	3	0.0306 (15625)	1.010 at dim 1331	0.0417	no
simplex (e)	4	0.0061 (64065)	1.032 at dim 495	1/80 = 0.0125	no
box (x)	4	0.0048 (390625)	1.002 at dim 83521	0.0125	no

- **[derived by computation] No lattice in the family exceeds the asymptote.** At equal dimension the best gain over the JM curve is **≤ 8 %**, at small dimension only. Above a few hundred it is ≤ 3 %. These are extended-strategy shifts, not a new exponent (Figure 4).
- **Reading.** This is consistent with F1–F4 and with L4-B's conditional closure. It is a finite family, not the LP, so P3 remains formally **open**. The ePrint's formula, read and run, is the only thing that could still change it.

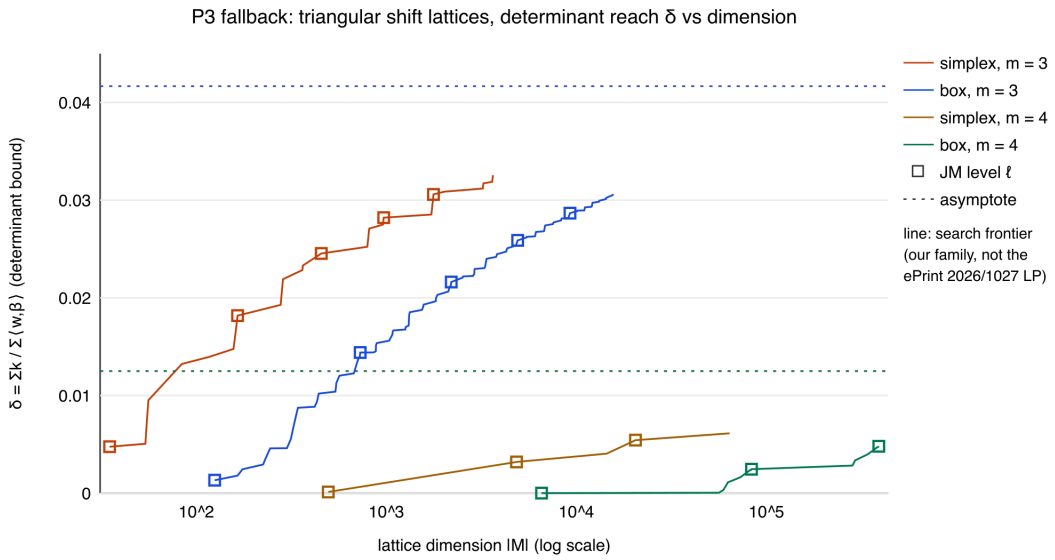


Figure 5: P3 fallback: δ against dimension for our shift-lattice family (lines) and the JM levels (squares), with the asymptotes. Source: p3/analysis/p3.svg.

7b.5 Where lead 4 stands

premise / probe	status
P1, irreducible over F_p	measured: GP and native, 20/20 at m = 3 and 4
P1, absolutely irreducible	not decidable by line restrictions; recalled (Diem)
P2, full box	measured: GP and native, dense, 20/20
P3, automated-Coppersmith LP	open; our finite search finds nothing past the JM asymptote
L4-A, measured reach	measured at m = 3, $\ell \le 2$: no success at $B \ge 2$, matching §3.2 (v1 false positive: global relations, amended)
L4-C, gap structure	measured: null, 41/41 cells inside the band
L4-D	not run

Lead 4 stays **closed for lattice oracles on coordinate factor bases**, conditional only on P3. No measured cost changed, so this is not an improvement under the success criterion, and no dashboard point is added (AGENTS §7a).

7c. Addendum (2026-10-29): a bound on every shift strategy, the P3 LP included

`research/smallroot_optimal_reach_20261026` (merged 2026-10-26) proves this for one polynomial `f` modulo `p` in `n` unknowns with equal bounds `|x_i| < p^δ`. Every Coppersmith lattice whose rows lie in `(p, f)^t`, on the lattice points of a convex monomial set, satisfies Howgrave-Graham only if

`ε = n·δ ≤ (n! · vol Newt(f))^{−1/n}`.

Its §2 gives the argument: regular sequence, Ostrowski, Brunn–Minkowski. It covers Jochemsz–May, Ryan's optimal shifts and any LP-chosen shift set, because all of them are sublattices of `L(M)`. The proof drops dimension factors, which favours the attack.

- **[derived] Both premises hold here.**
  - The small-x factor base gives every `x_i` the same bound `B`.
  - L4-B measured the Newton polytopes. `Newt(S₄(·; z))` is the full box `[0,4]³` (volume 64), and `Newt(S₅(·; z))` is the full box `[0,8]⁴` (volume 4096), on 20/20 values of `z` (§7a, §7b.1).
- **[derived] The bound.** The formula reproduces the study's S_3 value of 0.3293 (`vol = 14/3`, `n = 3`). For this note's targets it gives:

polynomial	n	vol Newt	ε bound	per-variable δ bound	target δ	short by
S_4 (m = 3)	3	64	0.1376	0.0459 = 1/21.8	1/4 (milestone)	5.5×
S_5 (m = 4)	4	4096	0.0565	0.0141 = 1/70.8	1/5 (goal)	14.2×

The bounds sit just above the Jochemsz–May asymptotes, 1/24 and 1/80, as an upper bound should. The P3 fallback's best values, 0.0306 at m = 3 and 0.0048 at m = 4 (x-box), sit below them.

What this closes. The "real P3 LP" left open in §0 and §7a cannot reach either target with x-variable lattices on convex monomial sets. That holds for every shift strategy, not only the triangular families searched in §7b.4.

What it does not close.

- **The symmetric e-form.** Its bounds `C(m,i)·B^i` are unequal, and for unequal bounds the theorem bounds only `∏ δ_i`. The cited study's numerical optimiser found nothing above 1/2 for S_3 there. §7b.4 found no e-form family above the JM asymptote. A proof for the e-form is still missing.
- **Non-convex monomial sets.**

- **Lattices that use more than the one ideal `(p, S_{m+1})^t`,** for example several targets' polynomials at once.

These are the remaining lattice items. They are listed in the cited study's §5 for S_3 and carry over here.

8. Sources

Each entry is marked with how far it was read.

- `[internal] RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` (2026-10-08): §2.1–2.4 cost model and JM table; Probe B/C tables; §4.6 closures.
- `[internal]` `RESEARCH_IC_LEADS_AND_CONNECTIONS_20261010.md` §8–§10 (worktree copy).
- `[kb] crypto-autoresearcher` frontier rows KR-IC-5931ee and KR-IC-5a43e5; KN-LIT-d8e1b9 (Yokoyama, Yasuda, Takahashi, Kogure, *J. Math. Cryptol.* 14 (2020) 460–485, doi:10.1515/jmc-2019-0029).
- `[retrieved, 2-page abstract read]` G. McGuire, D. Mueller, "A few more index calculus algorithms for the ECDLP", ePrint 2017/1262 (Caen 2018 abstract): O(p log² p), Gröbner-free. Its reference list confirms Amadori–Pintore–Sala ePrint 2017/609 and Faugère–Huot–Joux–Renault–Vitse, EUROCRYPT 2014, LNCS 8441, 40–57. https://nitaj.users.lmno.cnrs.fr/Caen2018/DanielaMuellerCaen2018.pdf
- `[retrieved, abstract]` J. Hu, "Implementation analysis of index calculus method on elliptic curves over prime finite fields", ePrint 2024/1923. https://eprint.iacr.org/2024/1923
- `[retrieved, abstract]` Y. Aono, M. Agrawal, T. Satoh, O. Watanabe, "On the optimality of lattices for the Coppersmith technique", ePrint 2012/108; ACISP 2012; AAECC 29(2) 2018. https://eprint.iacr.org/2012/108
- `[retrieved, abstract]` J. Xu, S. Sarkar, H. Wang, L. Hu, "Improving bounds on elliptic curve hidden number problem for ECDH key exchange", Asiacrypt 2022, ePrint 2022/1239. https://eprint.iacr.org/2022/1239
- `[retrieved, search abstract]` K. Meers, J. Nowakowski, automated Coppersmith, Asiacrypt 2023, ePrint 2023/1409. https://eprint.iacr.org/2023/1409
- `[retrieved, abstract]` Z. Ding, Z. Dai, B. Wu, Y. Zhang, K. Zhang, "Computing asymptotic bounds for the automated Coppersmith method via linear programming", ePrint 2026/1027. https://eprint.iacr.org/2026/1027
- `[retrieved, search abstract]` M.-C. Chang, J. Cilleruelo, M. Garaev, J. Hernández, I. Shparlinski, A. Zumalacárregui, "Points on curves in small boxes and applications", arXiv:1111.1543. https://arxiv.org/abs/1111.1543
- `[recalled]` I. Semaev, ePrint 2004/031. E. Jochemsz, A. May, Asiacrypt 2006. D. Coppersmith, J. Cryptology 1997. Bostan–Gaudry–Schost, SIAM J. Comput. 2007. Ostrowski's theorem on Newton polytopes of products. Jacobson–Koblitz–Silverman–Stein–Teske, DCC 2000 (xedni). Petit–Kosters–Messeng, PKC 2016. Gaudry–Thériault–Thomé–Diem (genus-g double large prime).
- **Searches that found nothing beyond the above** (2026-10-28):

- “summation polynomials prime field … Coppersmith small roots lattice index calculus”;
- “ePrint 2023 2024 2025 index calculus elliptic curves prime field summation polynomial”;
- “elliptic curve hidden number problem x-coordinate Coppersmith 2020–2023”.

None of them turned up a small-root method for summation polynomials past the JM reach, or any sub-√p prime-field algebraic ECDLP result. Hu 2024 states that no subexponential method is known.

9. Files and rendering

- `README.md`: this report.
- `delta_arity_map.svg`: Figure 1, hand-written and editable. Every coordinate follows the formula in its header comment.
- `delta_arity_map.png`: the figure rendered with `rsvg-convert -w 2160 delta_arity_map.svg -o delta_arity_map.png`.
- `report.pdf`: rendered from this file with every figure link pointed at its PNG (`pandoc … --pdf-engine=typst`), after §7b was added.
- `l4b/`, `l4c/`, `l4a/`, `p3/`: one directory per §7b probe. Each holds `PROTOCOL.md` (with `AMENDMENT-1.md` for L4-A), `REGISTRATION.sha256`, `BINARY.sha256` (binary and source hashes), raw `data/`, `derive.mjs` (a thin Node derivation over the raw data; tooling, not research code) and `analysis/` (tables, editable SVG figures, PNG renders made with `rsvg-convert -w 2160`).
- The native code lives outside this directory: `src/cryptanalysis/semaev_prime_poly.rs` and the four `examples/semaev_*.rs` programs named in §7b.